

MID EXAM

Course: Data structures (cs01)

Set: B | Time: 1 Hour | Max Marks: 41 | Date: 14-03-2026

Total Questions: 9 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. Which of the following accurately distinguishes an architectural pattern from a reference model based on the provided text? [4 Marks]

- a) An architectural pattern is an architecture, whereas a reference model is a division of functionality.
- b) An architectural pattern describes element and relation types with constraints, while a reference model outlines functionality.
- c) A reference model is the architect's first major design choice, while an architectural pattern is characteristic of mature domains.
- d) Architectural patterns always exhibit known quality attributes, but reference models are only concerned with decomposition.

2. According to the definition of software architecture provided, what is a key implication regarding the level of detail it represents? [4 Marks]

- a) Architecture must capture all granular implementation details of every element for completeness.
- b) It defines software elements and their relationships but also suppresses details not affecting how elements interact.
- c) An architecture primarily focuses on public interfaces and internal element logic without abstraction.
- d) It dictates a single, dominant structure for the system, making other structures secondary.

3. When considering the properties of modules in a Module View, which of the following is NOT listed as a potential property to guide implementation or analysis? [4 Marks]

- a) Revision history
- b) Visibility of interface(s)
- c) Performance characteristics
- d) Responsibilities

4. In the context of Component-and-Connector (C&C) views, which property would an architect most likely use to determine if a proposed hardware configuration is adequate? [4 Marks]

- a) Modifiability
- b) Functionality
- c) Resource requirements
- d) Concurrency

SECTION A: SHORT ANSWERS

5. Architects require more than just technical skills due to stakeholder engagement. Explain how stakeholders influence architectures and what non-technical skills are considered essential for architects to manage this influence effectively. [5 Marks]

6. Describe the core idea of an "Exception or error-handling view" and elaborate on its specific utility for understanding a system's architecture. **[5 Marks]**
7. Contrast the "Deployment" and "Implementation" types of Allocation Structures, highlighting what each view specifically maps. **[5 Marks]**
8. A "Quality attribute scenario" is a key tool for describing quality attribute requirements. Explain its purpose and list the six essential parts it consists of. **[5 Marks]**
9. During the architecture design process, creating a "skeletal system" is recommended. Explain the primary idea behind constructing a skeletal system and describe its initial steps. **[5 Marks]**
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